Namma Space: Capture SOP (Standard Operating Procedure)

Goal: a non-technical person captures an indoor space with only a smartphone, and the pipeline (`namma build`) turns it into a metric-scale 3D Gaussian Splat with a walkable navigation grid. **Inputs accepted:** videos (`.mp4 .mov .m4v .mkv .3gp ...`) and/or photos (`.jpg .png`) in one folder; many clips of the same space are merged automatically. **Equipment:** any smartphone, plus (recommended) 3-4 printed ArUco markers. No LiDAR, tripod or other hardware.

1. Before you start (10 min)

- ☐ **All lights on**; diffuse daylight is fine. Avoid direct sunlight patches and glare on floors and screens.
- ☐ **Clear clutter from the floor** so the floor stays visible; the floor drives scale and the walkable grid.
- ☐ **Close doors you do not want scanned.** Avoid or cover **mirrors and glass**; reflections create ghost geometry and break pose estimation.
- ☐ Keep people, pets and fans out of the shot. Moving objects become floaters.
- ☐ Clean the lens, charge the phone (>50%), free 5 GB of storage.
- ☐ Record in **portrait or landscape consistently** (do not switch inside one clip). Lock exposure/focus if the camera app allows it; use 1080p or 4K at 30 fps.
- ☐ **Metric scale, option A (preferred): ArUco markers.** Print 3-4 markers (dictionary `DICT_4X4_50` , ids 0-3) at exactly **15 cm** side length (pipeline default `--marker-size 0.15`). Tape them flat on the floor, spread across the space, each visible from several viewpoints.
- ☐ **Metric scale, option B: phone-height SOP.** With no markers, hold the phone at a constant **1.35 m** above the floor (about chest height; pipeline default `--capture-height 1.35`) for the whole capture. Scale is derived from that height, so keep it steady.

2. Capture pattern (about 3-5 min per room)

- ☐ **Walk slowly** (about one step per second); never spin in place. Fast turns blur frames and break SfM.
- ☐ **Loop 1, perimeter:** walk the full perimeter, camera facing the opposite walls and the centre.
- ☐ **Loop 2, inner pass:** walk a second loop or lawn-mower pattern through the middle, looking across and slightly down at the floor.
- ☐ **Corner shots:** stand 1-1.5 m from every corner and point the phone into it, sweeping up and down.
- ☐ **Ceiling and floor sweeps:** tilt up to the ceiling edge and down to the floor along each wall.
- ☐ Keep **60-70% overlap** between consecutive views; always keep part of the previous view in frame.
- ☐ Photograph furniture and POI items (reception, signboards, counters) from 2-3 sides as extra close-up photos.
- ☐ Do not film only a blank wall; include edges, corners and objects so SfM has features.

3. Doorways and multiple rooms

- ☐ **Small space:** one continuous video through all rooms. Slow down at every doorway; look back into the room just left, then forward into the next, so frames overlap across the threshold.
- ☐ **Large space:** one clip per zone, each starting and ending in an area already covered by a neighbouring clip (about 2 m overlap). Name clips in order (`01_entrance.mp4` , `02_hall.mp4`).
- ☐ Keep doors **fully open and propped**.
- ☐ Walk the **actual routes people will use**: the pipeline treats cells you walked through as free, which repairs gaps on plain floors.

4. After capture

- ☐ Put all videos and photos for one site in a single folder.

- [ ] Run `namma build --input <folder> --work <workdir> --gsplat-dir <gsplat> --out <scene> --name "<Site>"`.
- [ ] Check the build report: frames registered, markers found, scale source (`markers` or `camera-height`), PSNR, walkable area.
- [ ] If too few frames register, or the "capture too short or too blurry" error appears, re-record the weak area with more overlap and slower motion.
- [ ] **Alternative route:** capture with Polycam, Luma AI or Scaniverse and export a `.ply` splat. Process it with `namma process-splat --splat <file.ply> --out <scene>` (the splat must already be metric and floor-at-zero; pass `--cameras` if walked positions are available).

5. Quick quality gate

Check	Pass condition
Coverage	Every wall, corner, floor area and ceiling edge seen from several viewpoints
Sharpness	No blurred stretches (pipeline drops frames below 35% of median sharpness)
Lighting	No blown-out windows, no very dark rooms
Artifacts	No mirror/glass ghosting; few floaters after cleaning
Scale	Markers detected (3+ views each) or constant 1.35 m phone height
Output	`scene.spz`, `scene.json` and navgrid load in the web viewer